6	In a photoelectric experiment, an ultraviolet (UV) light source of constant intensity and single frequency is used. Two metal plates X and Y are contained in an evacuated glass container and are connected to a circuit as shown in Fig. 6.1. The UV source is placed at a distance away from X and Y.	

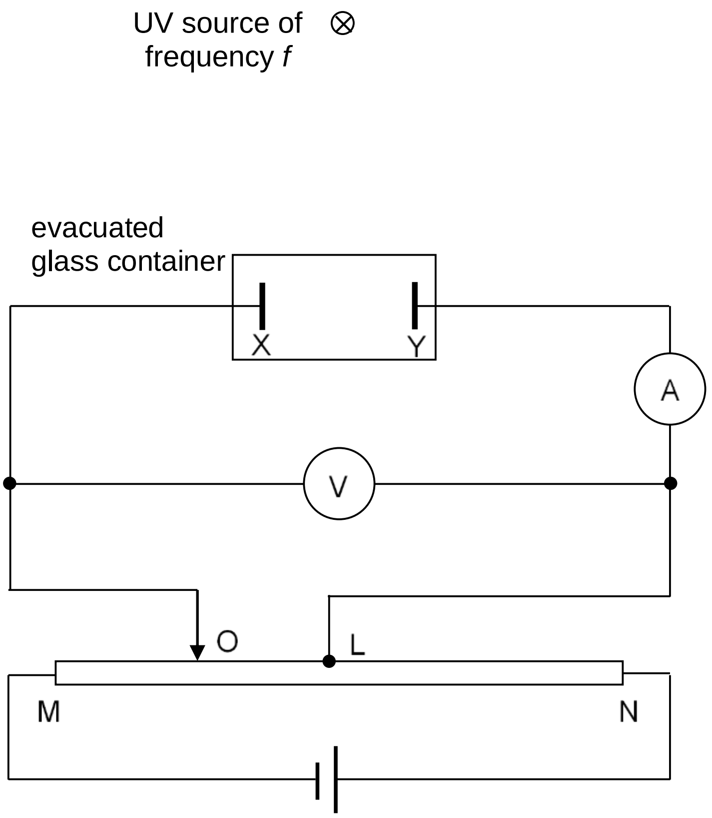


Fig. 6.1

The graph shown in Fig. 6.2 depicts the relationship between the voltmeter reading and the ammeter reading when metal plate X is the photoelectric emitter. No photoelectrons are emitted from Y.

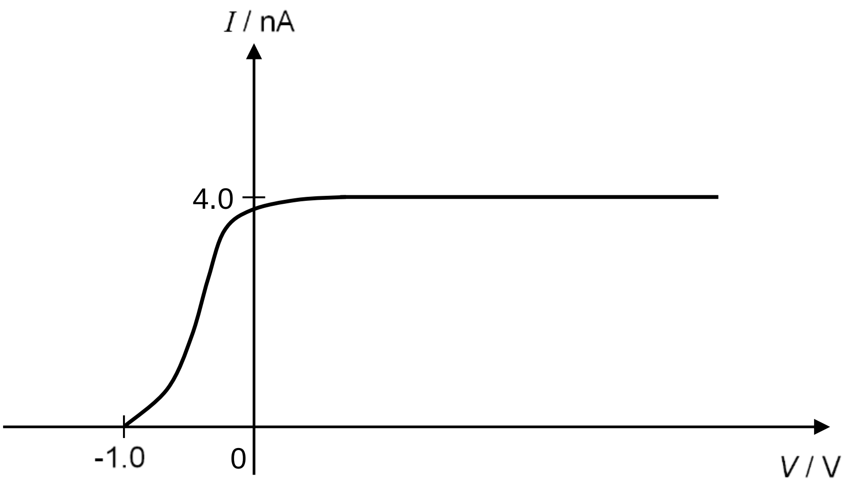
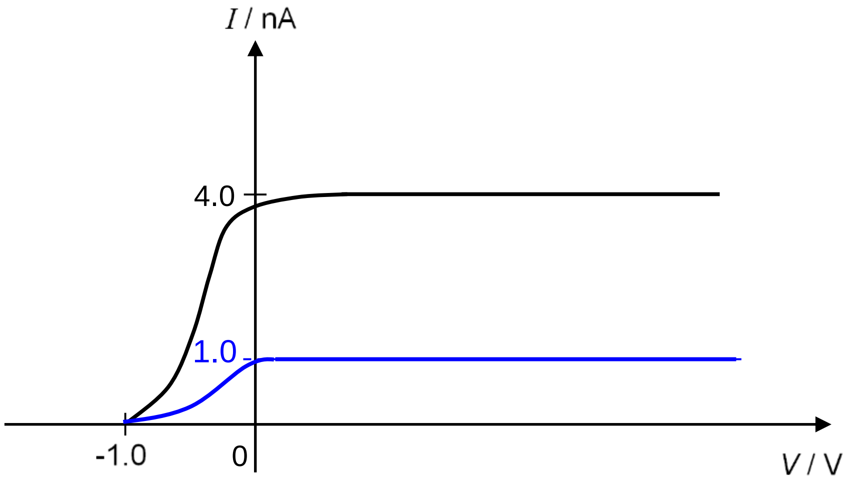
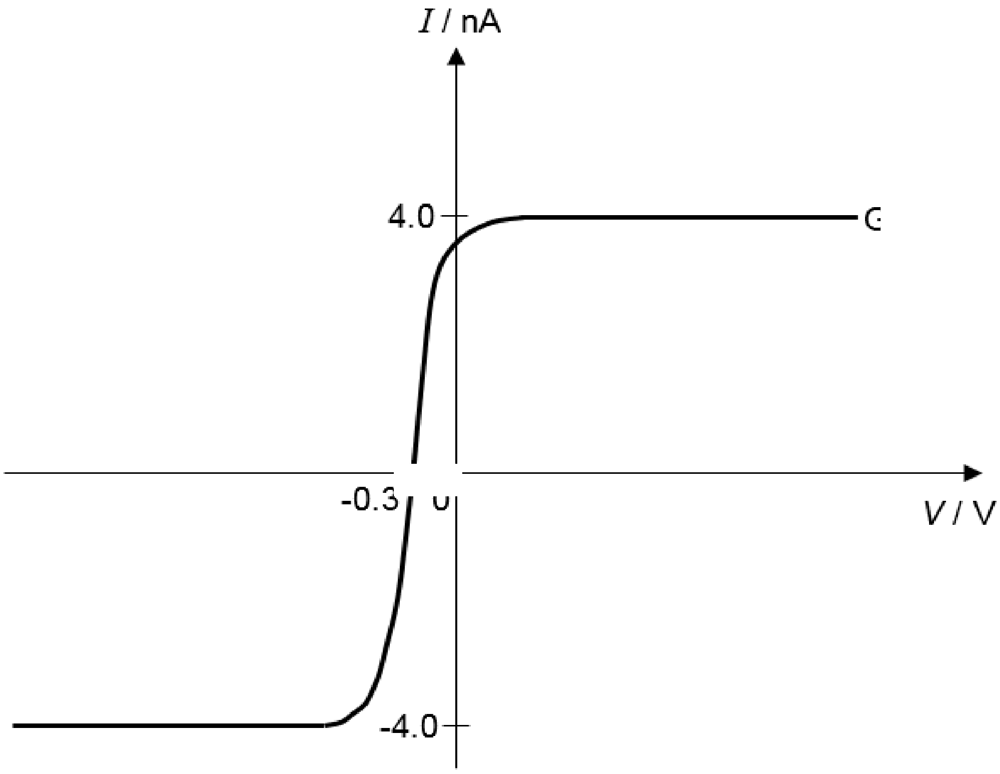


Fig. 6.2

	(a)	Explain why the current remains constant for positive values of V.
		....

		………	
		………	
		………	
		………	[3]
		The **current measured is proportional to the rate of electrons reaching Y.** **The rate of <u>electron</u> ejection from X arriving at Y is fixed because it is proportional to the rate of <u>photon</u> incidence on X which is fixed when electromagnetic radiation is at a fixed intensity and fixed frequency.** **For positive values of *V*, the electrons experience an accelerating force towards Y. Thus, ALL the electrons ejected from X will be collected at Y regardless of the applied potential difference.** Therefore, the current will also be constant at a maximum value.	**1** **1** **1**
	(b)	Metal plate X is made of zinc with a work function of 3.8 eV. Using information from Fig. 6.2, determine the wavelength of the UV source.	
		wavelength = ……………………………. m	[2]
		Applying the photoelectric equation, Photon Energy = Work function of metal surface + Maximum K.E. of emitted electrons $\frac{hc}{\lambda} = \phi + eV_s$ $\lambda = \frac{hc}{\phi + eV_s}$ $= \frac{6.63\times10^{-34}(3.00\times10^{8})}{3.8(1.60\times10^{-19}) + 1.60\times10^{-19}(1.0)}$ $= 2.59\times10^{-7}$ m	 **1** **1**
	(c)	Sketch, on Fig. 6.2, the graph when the experiment was repeated with UV light source of the same frequency but with intensity one-quartered.	[2]
		LHS of graph: Same frequency photons incident and same work function energy of the metal ➔ Maximum K.E. of electrons ejected unchanged ➔ **Stopping potential unchanged** RHS of graph: $Intensity\ of\ radiation = \frac{P}{A} = \frac{E}{tA} = \frac{Nhf}{tA}$ $\frac{N}{t} = Intensity\ of\ radiation\ x\ \frac{A}{hf}$ Same frequency f and Intensity one-quartered ➔ No. of photons incident per unit time, $\frac{N}{t}$ will be one-quarter the original. ➔ Since each electron ejected is the result of absorbing one photon, the photoelectric current is proportional to $\frac{N}{t}$. Thus, **maximum photoelectric current will be <u>one-quarter</u> the original.**	

			**1** **1**
	(d)	The UV light source was replaced with another light source of higher frequency. The graph in Fig. 6.3 was obtained when the experiment was conducted using the higher frequency light source.  **Fig. 6.3**	
		Explain which metal plate, X or Y, has a greater work function energy.	
		…………………………………………………………………………………………………	
		…………………………………………………………………………………………………	[2]
		Fig. 6.3 shows that photoelectron emission occurs in <u>both</u> metal plates X and Y when light source of higher frequency is used. Energy per photon is proportional to its frequency, and for an electron to be ejected from a metal surface the photon energy must be greater or equal to the work function energy of the metal surface.	**1**

		Since Y emits electrons only with the higher frequency light source but X emits electrons with both the lower and higher frequency light source, Y must have the greater work function energy.	**1**

7	**(a)**	State experimental evidence to suggest that the process of radioactive decay is